LimitEye: Overspeeding Detection and Automated Penalty System Using Machine Learning

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Abstract

A brief overview of the project:

- Goal: To develop an AI-based system for detecting vehicle overspeeding and automatically issuing fines.
- Core Idea: Using a custom-trained YOLO model and a tracking algorithm (SORT), the system analyzes live video feeds, calculates vehicle speed, and sends fine notifications via email to vehicle owners.
- Impact: Ensures road safety and enforces speed limits effectively.

Introduction

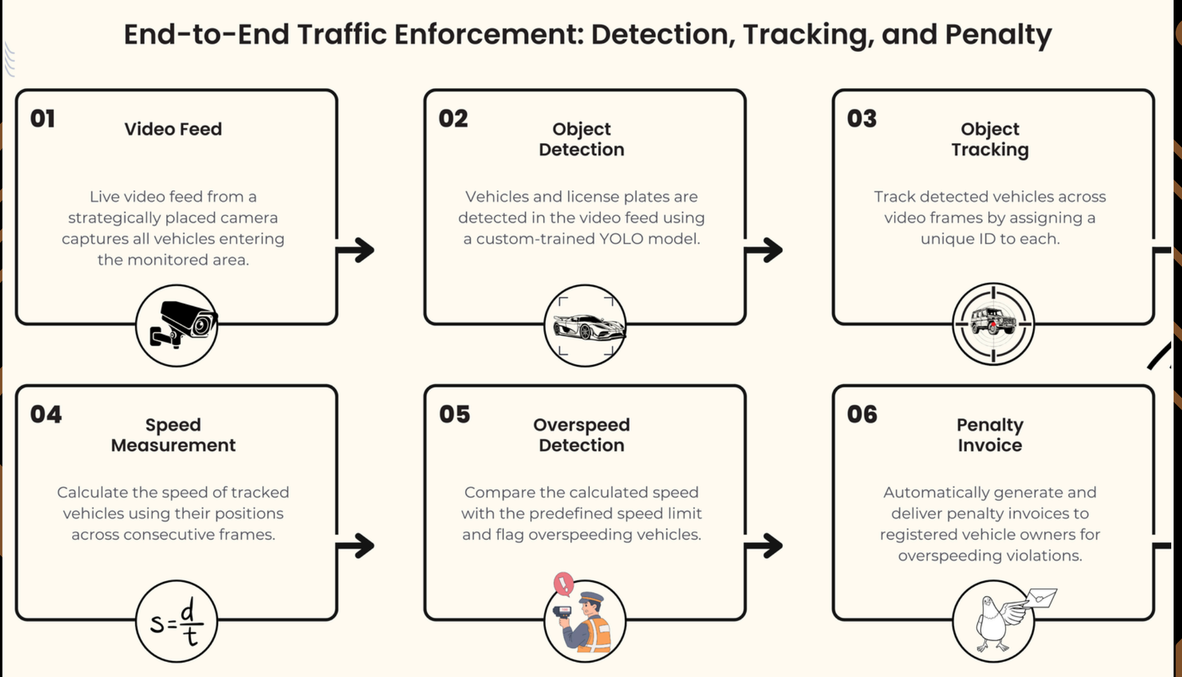
Problem Statement:

- Speeding is a major cause of traffic accidents. Traditional speed detection systems are expensive and limited in functionality.

Objective:

- To design an affordable and scalable solution for real-time speed detection and automated fine issuance.

Methodology



Tools Used

ChatGPT

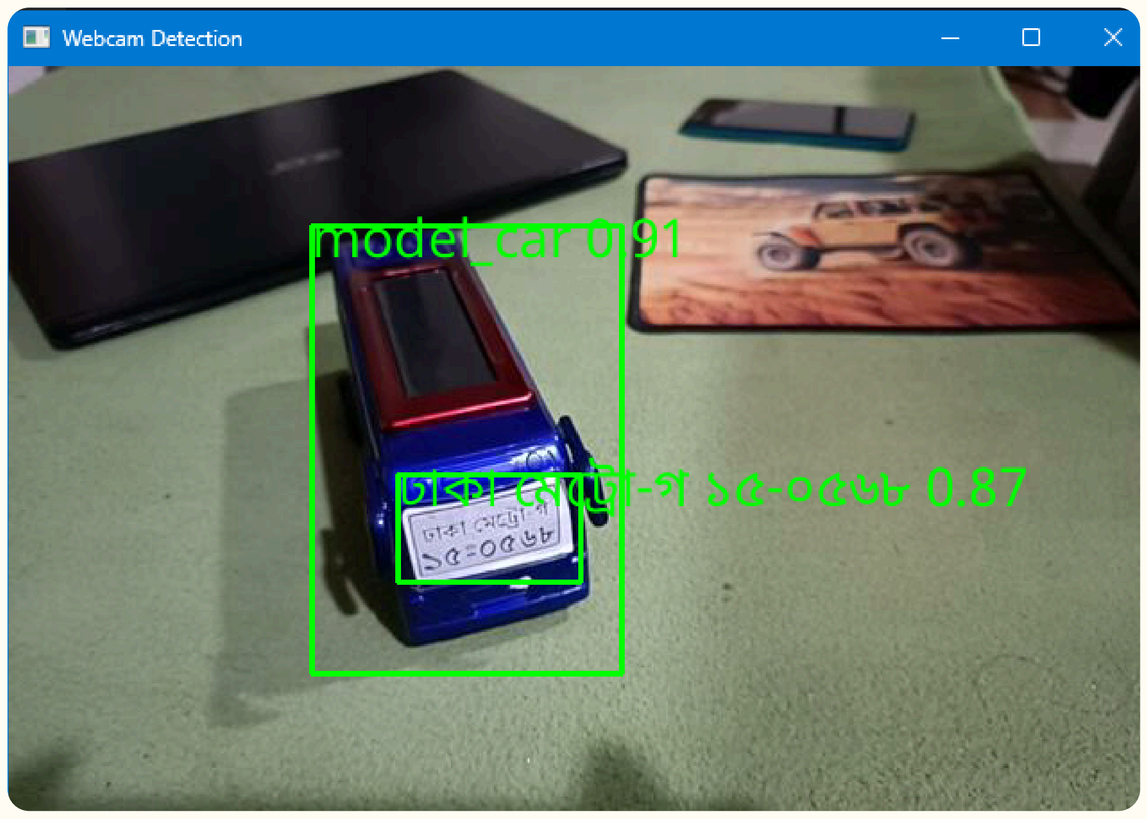
LabelImg

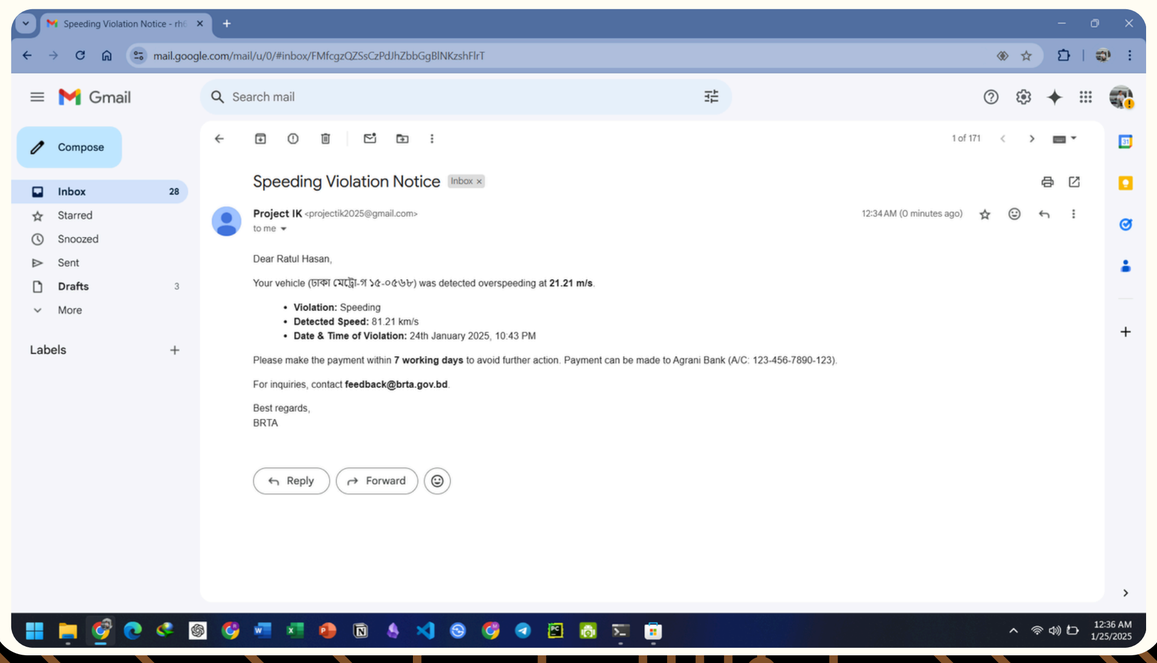
Google Colab

Used Models

- YOLO Model: Trained with custom dataset for vehicle and license plate detection.
- SORT Algorithm: Tracks vehicle movement efficiently.
- Email Automation: Fine notifications sent using SMTP.

Results





Future Work

- Enhance detection in adverse weather using thermal imaging and preprocessing.
- Expand vehicle classification to include motorcycles and heavy vehicles.
- Integrate with national databases for automated owner identification.
- Scale for deployment in smart city traffic systems.
- Optimize AI models for faster, more efficient detection.
- Implement privacy safeguards to balance enforcement and data security.

Conclusion

LimitEye demonstrates a practical, AI-driven solution for real-time speed monitoring and automated fine issuance. By streamlining traffic enforcement, it enhances road safety and reduces manual intervention. With its scalability and precision, LimitEye holds the potential to become a cornerstone in modern traffic management systems.

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